

## Q1 2021 (01 Sep Shift 2)

Let  $a_1, a_2, \dots, a_{21}$  be an  $AP$  such that  $\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} = \frac{4}{9}$ .

If the sum of this  $AP$  is 189, then  $a_{66}$  is equal to :

(1) 57

(2) 72

(3) 48

(4) 36

## Q2 2021 (01 Sep Shift 2)

Let  $S_n = 1 \cdot (n-1) + 2 \cdot (n-2) + 3 \cdot (n-3) + \dots + (n-1) \cdot 1, n \geq 4$

The sum  $\sum_{n=4}^{\infty} \left( \frac{2S_n}{n!} - \frac{1}{(n-2)!} \right)$  is equal to :

(1)  $\frac{e-1}{3}$

(2)  $\frac{e-2}{6}$

(3)  $\frac{e}{3}$

(4)  $\frac{e}{6}$

## Q3 2021 (31 Aug Shift 2)

If  $S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + \dots$ , then  $160S$  is equal to .

## Q4 2021 (31 Aug Shift 2)

Let  $a_1, a_2, a_3, \dots$  be an  $A.P.$  If

$\frac{a_1 + a_2 + \dots + a_{10}}{a_1 + a_2 + \dots + a_p} = \frac{100}{p^2}, p \neq 10$ , then  $\frac{a_{11}}{a_{10}}$  is equal to:

(1)  $\frac{19}{21}$

(2)  $\frac{100}{121}$

(3)  $\frac{21}{19}$

(4)  $\frac{121}{100}$

## Q5 2021 (31 Aug Shift 1)

Three numbers are in an increasing geometric progression with common ratio  $r$ .

If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference  $d$ .

If the fourth term of GP is  $3r^2$ , then  $r^2 - d$  is equal to :

(1)  $7 - 7\sqrt{3}$

(2)  $7 + \sqrt{3}$

(3)  $7 - \sqrt{3}$

(4)  $7 + 3\sqrt{3}$

## Q6 2021 (31 Aug Shift 1)

The sum of 10 terms of the series

$$\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \dots \text{ is :}$$

(1) 1

(2)  $\frac{120}{121}$

(3)  $\frac{99}{100}$

(4)  $\frac{143}{144}$

## Q7 2021 (27 Aug Shift 2)

If  $0 < x < 1$  and  $y = \frac{1}{2}x^2 + \frac{2}{3}x^3 + \frac{3}{4}x^4 + \dots$ , then the value of  $e^{1+y}$  at  $x = \frac{1}{2}$  is:

(1)  $\frac{1}{2}e^2$

(2)  $2e$

(3)  $\frac{1}{2}\sqrt{e}$

(4)  $2e^2$

## Q8 2021 (27 Aug Shift 1)

If  $0 < x < 1$ , then  $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots$ , is equal to:

$$(1) x \left( \frac{1+x}{1-x} \right) + \log_e(1-x)$$

$$(2) x \left( \frac{1-x}{1+x} \right) + \log_e(1-x)$$

$$(3) \frac{1-x}{1+x} + \log_e(1-x)$$

$$(4) \frac{1+x}{1-x} + \log_e(1-x)$$

**Q9 2021 (26 Aug Shift 2)**

Let  $a_1, a_2, \dots, a_{10}$  be an  $AP$  with common difference  $-3$  and  $b_1, b_2, \dots, b_{10}$  be a  $GP$  with common ratio  $2$ . Let  $c_k = a_k + b_k, k = 1, 2, \dots, 10$ . If  $c_2 = 12$  and  $c_3 = 13$ , then  $\sum_{k=1}^{10} c_k$  is equal to .

**Q10 2021 (26 Aug Shift 1)**

If the sum of an infinite GP  $a, ar, ar^2, ar^3, \dots$  is  $15$  and the sum of the squares of its each term is  $150$ , then the sum of  $ar^2, ar^4, ar^6, \dots$  is :

$$(1) \frac{5}{2}$$

$$(2) \frac{1}{2}$$

$$(3) \frac{25}{2}$$

$$(4) \frac{9}{2}$$

**Q11 2021 (26 Aug Shift 1)**

The sum of the series

$$\frac{1}{x+1} + \frac{2}{x^2+1} + \frac{2^2}{x^4+1} + \dots + \frac{2^{100}}{x^{2^{100}}+1} \text{ when } x = 2 \text{ is:}$$

$$(1) 1 + \frac{2^{101}}{4^{101}-1}$$

$$(2) 1 + \frac{2^{100}}{4^{101}-1}$$

$$(3) 1 - \frac{2^{100}}{4^{100}-1}$$

$$(4) 1 - \frac{2^{101}}{4^{101}-1}$$

# Answer Key

Q1 (2)

Q2 (1)

Q3 (305)

Q4 (3)

Q5 (2)

Q6 (2)

Q7 (1)

Q8 (1)

Q9 (2021)

Q10 (2)

Q11 (4)

Q1 (2)

$$\begin{aligned}\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} &= \sum_{n=1}^{20} \frac{1}{a_n (a_n + d)} \\ &= \frac{1}{d} \sum_{n=1}^{20} \left( \frac{1}{a_n} - \frac{1}{a_n + d} \right) \\ \Rightarrow \frac{1}{d} \left( \frac{1}{a_1} - \frac{1}{a_{21}} \right) &= \frac{4}{9} \text{ (Given)} \\ \Rightarrow \frac{1}{d} \left( \frac{a_{21} - a_1}{a_1 a_{21}} \right) &= \frac{4}{9} \\ \Rightarrow \frac{1}{d} \left( \frac{a_1 + 20d - a_1}{a_1 a_2} \right) &= \frac{4}{9} \Rightarrow a_1 a_2 = 45 \dots (1)\end{aligned}$$

$$\text{Now sum of first 21 terms} = \frac{21}{2} (2a_1 + 20d) = 189$$

$$\Rightarrow a_1 + 10d = 9 \dots (2)$$

For equation (1) &amp; (2) we get

$$a_1 = 3 \text{ \& } d = \frac{3}{5}$$

OR

$$a_1 = 15 \text{ \& } d = -\frac{3}{5}$$

$$\text{So, } a_6 \cdot a_{16} = (a_1 + 5d)(a_1 + 15d)$$

$$\Rightarrow a_6 a_{16} = 72$$

Option (2)

Q2 (1)

$$\text{Let } T_r = r(n - r)$$

$$T_r = nr - r^2$$

$$\Rightarrow S_n = \sum_{r=1}^n T_r = \sum_{r=1}^n (nr - r^2)$$

$$S_n = \frac{n \cdot (n)(n+1)}{2} - \frac{n(n+1)(2n+1)}{6}$$

$$S_n = \frac{n(n-1)(n+1)}{6}$$

$$\text{Now } \sum_{r=4}^{\infty} \left( \frac{2S_n}{n!} - \frac{1}{(n-2)!} \right)$$

$$= \sum_{r=4}^{\infty} \left( 2 \cdot \frac{n(n-1)(n+1)}{6 \cdot n(n-1)(n-2)!} - \frac{1}{(n-2)!} \right)$$

$$= \sum_{r=4}^{\infty} \left( \frac{1}{3} \left( \frac{n-2+3}{(n-2)!} \right) - \frac{1}{(n-2)!} \right)$$

$$= \sum_{r=4}^{\infty} \frac{1}{3} \cdot \frac{1}{(n-3)!} = \frac{1}{3} (e - 1)$$

Option (1)

Q3 (305)

$$S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + \dots$$

$$\frac{1}{5} S = \frac{7}{5^2} + \frac{9}{5^3} + \frac{13}{5^4} + \dots$$

On subtracting

$$\frac{4}{5} S = \frac{7}{5} + \frac{2}{5^2} + \frac{4}{5^3} + \frac{6}{5^4} + \dots$$

$$S = \frac{7}{4} + \frac{1}{10} \left( 1 + \frac{2}{5} + \frac{3}{5^2} + \dots \right)$$

$$S = \frac{7}{4} + \frac{1}{10} \left( 1 - \frac{1}{5} \right)^{-2}$$

$$= \frac{7}{4} + \frac{1}{10} \times \frac{25}{16} = \frac{61}{32}$$

$$\Rightarrow 160 S = 5 \times 61 = 305$$

Q4 (3)

$$\frac{\frac{10}{2}(2a_1 + 9d)}{\frac{p}{2}(2a_1 + (p-1)d)} = \frac{100}{p^2}$$

$$(2a_1 + 9d)p = 10(2a_1 + (p-1)d)$$

$$9dp = 20a_1 - 2pa_1 + 10d(p-1)$$

$$9p = (20 - 2p)\frac{a_1}{d} + 10(p-1)$$

$$\frac{a_1}{d} = \frac{(10-p)}{2(10-p)} = \frac{1}{2}$$

$$\therefore \frac{a_{11}}{a_{10}} = \frac{a_1 + 10d}{a_1 + 9d} = \frac{\frac{1}{2} + 10}{\frac{1}{2} + 9} = \frac{21}{19}$$

Q5 (2)

Let numbers be  $\frac{a}{r}, a, ar \rightarrow G.P$ 

$$\frac{a}{r}, 2a, ar \rightarrow A.P \Rightarrow 4a = \frac{a}{r} + ar \Rightarrow r + \frac{1}{r} = 4$$

$$r = 2 \pm \sqrt{3}$$

$$4^{\text{th}} \text{ form of } G.P = 3r^2 \Rightarrow ar^2 = 3r^2 \Rightarrow a = 3$$

$$r = 2 + \sqrt{3}, a = 3, d = 2a - \frac{a}{r} = 3\sqrt{3}$$

$$r^2 - d = (2 + \sqrt{3})^2 - 3\sqrt{3}$$

$$= 7 + 4\sqrt{3} - 3\sqrt{3}$$

$$= 7 + \sqrt{3}$$

Q6 (2)

$$\begin{aligned}
 S &= \frac{2^2-1^2}{1^2 \times 2^2} + \frac{3^2-2^2}{2^2 \times 3^2} + \frac{4^2-3^2}{3^2 \times 4^2} + \dots \\
 &= \left[ \frac{1}{1^2} - \frac{1}{2^2} \right] + \left[ \frac{1}{2^2} - \frac{1}{3^2} \right] + \left[ \frac{1}{3^2} - \frac{1}{4^2} \right] + \dots + \left[ \frac{1}{10^2} - \frac{1}{11^2} \right] \\
 &= 1 - \frac{1}{121} \\
 &= \frac{120}{121}
 \end{aligned}$$

Q7 (1)

$$\begin{aligned}
 y &= \left(1 - \frac{1}{2}\right)x^2 + \left(1 - \frac{1}{3}\right)x^3 + \dots \\
 &= (x^2 + x^3 + x^4 + \dots) - \left(\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots\right) \\
 &= \frac{x^2}{1-x} + x - \left(x + \frac{x^2}{2} + \frac{x^2}{3} + \dots\right) \\
 &= \frac{x}{1-x} + \ln(1-x) \\
 x = \frac{1}{2} &\Rightarrow y = 1 - \ln 2 \\
 e^{1+y} &= e^{1+1-\ln 2} \\
 &= e^{2-(\ln 2)} = \frac{e^2}{2}
 \end{aligned}$$

Q8 (1)

$$\begin{aligned}
 \text{Let } t &= \frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots \infty \\
 &= \left(2 - \frac{1}{2}\right)x^2 + \left(2 - \frac{1}{3}\right)x^3 + \left(2 - \frac{1}{4}\right)x^4 + \dots \infty \\
 &= 2(x^2 + x^3 + x^4 + \dots \infty) - \left(\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty\right) \\
 &= \frac{2x^2}{1-x} - (\ln(1-x) - x) \\
 \Rightarrow t &= \frac{2x^2}{1-x} + x - \ln(1-x) \\
 \Rightarrow t &= \frac{x(1+x)}{1-x} - \ln(1-x)
 \end{aligned}$$

Q9 (2021)

$$\begin{aligned}
 c_2 &= a_2 + b_2 = a_1 - 3 + 2b_1 = 12 \\
 a_1 + 2b_1 &= 15 \quad (1) \\
 c_3 &= a_3 + b_3 = a_1 - 6 + 4b_1 = 13
 \end{aligned}$$

$$a_1 + 4b_1 = 19 \quad (2)$$

$$\text{from (1) \& (2) } b_1 = 2, a_1 = 11$$

$$\sum_{k=1}^{10} c_k = \sum_{k=1}^{10} (a_k + b_k) = \sum_{k=1}^{10} a_k + \sum_{k=1}^{10} b_k$$

$$= \frac{10}{2}(2 \times 11 + 9 \times (-3)) + \frac{2(2^{10}-1)}{2-1}$$

$$= 5(22 - 27) + 2(1023)$$

$$= 2046 - 25 = 2021$$

**Q10 (2)**

Sum of infinite terms :

$$\frac{a}{1-r} = 15 \dots\dots (i)$$

Series formed by square of terms:

$$a^2, a^2r^2, a^2r^4, a^2r^6 \dots\dots$$

$$\text{Sum} = \frac{a^2}{1-r^2} = 150$$

$$\Rightarrow \frac{a}{1-r} \cdot \frac{a}{1+r} = 150 \Rightarrow 15 \cdot \frac{a}{1+r} = 150$$

$$\Rightarrow \frac{a}{1+r} = 10 \dots\dots (ii)$$

$$\text{by (i) and (ii) } a = 12; r = \frac{1}{5}$$

Now series :  $ar^2, ar^4, ar^6$

$$\text{Sum} = \frac{ar^2}{1-r^2} = \frac{12 \cdot (1/25)}{1-1/25} = \frac{1}{2}$$

**Q11 (4)**

$$S = \frac{1}{x+1} + \frac{2}{x^2+1} + \frac{2^2}{x^4+1} + \dots + \frac{2^{100}}{x^{2^{100}}+1}$$

$$S + \frac{1}{1-x} = \frac{1}{1-x} + \frac{1}{x+1} + \dots = \frac{2}{1-x^2} + \frac{2}{1+x^2} + \dots$$

$$S + \frac{1}{1-x} = \frac{2^{101}}{1-x^{2^{101}}}$$

Put  $x = 2$

$$S = 1 - \frac{2^{101}}{2^{2^{101}} - 1}$$

Not in option (BONUS)